

Bluestock Mutual Fund Analytics Platform

Capstone Project Report

Full-stack ETL, Analytics, and Visualization Pipeline
for Indian Mutual Fund Data

July 2026

1. Executive Summary

This report presents the design, implementation, and results of a full-stack Mutual Fund Analytics Platform built as part of the Bluestock capstone project. The platform ingests mutual fund data from public sources, transforms it through a Python ETL pipeline into a SQLite star schema, and exposes insights via Jupyter notebooks and an interactive Power BI dashboard.

Project Overview

Project: Full-stack Mutual Fund Analytics Platform

Data Sources: AMFI India (NAV data), mfapi.in (meta, schemes, transactions)

Pipeline: Python ETL -> SQLite Star Schema -> Jupyter Notebooks (EDA, Performance, Advanced) -> Power BI Dashboard

Duration: 2022-2026 (4 years of monthly NAV data across ~40 schemes)

Key Findings

- SBI Bluechip Fund leads all schemes in Assets Under Management (AUM).
- Small Cap funds exhibit the highest absolute returns but also carry the highest Value-at-Risk (VaR).
- UPI is the dominant transaction mode, accounting for over 40% of investor transactions.
- Investors aged 36-45 form the most active demographic segment.
- SBI Mutual Fund dominates the industry with approximately 6.05 lakh crore in AUM.

Deliverables

- Production-grade ETL pipeline with cron-ready scheduling
- SQLite database with star-schema modelling (6 dimension/fact tables)
- 3 Jupyter notebooks covering EDA, Performance Analytics, and Advanced Analytics
- 4-page Power BI dashboard with slicers and drill-through
- Streamlit multi-page web app (Overview, Screener, Performance, Optimizer, Monte Carlo)
- Monte Carlo simulator for 5-year NAV projections with confidence bands
- Markowitz Efficient Frontier for optimal portfolio allocation
- Automated HTML email report generator

2. Data Pipeline & Schema

ETL Pipeline

The ETL pipeline is implemented in `scripts/etl_pipeline.py`. It ingests 10 raw CSV files containing mutual fund NAV history, scheme metadata, AUM data, transactions, and SIP data. The pipeline performs the following steps:

- Read raw CSVs from the `data/` directory using pandas.
- Clean and standardize columns: forward-fill (ffill) missing NAV values for weekends and holidays.
- Remove duplicate scheme entries and normalize date formats to ISO 8601.
- Strip unit suffixes (e.g., ' Cr', ' Lakh') from numeric columns and convert to float.
- Load processed data into a SQLite database (`mutual_fund.db`) following a star-schema design.

Star Schema Design

The database uses a classic star schema with the following tables:

<code>dim_fund</code>	- Fund attributes (scheme name, category, fund house, risk level)
<code>dim_date</code>	- Date dimension (date, year, quarter, month, day, weekday)
<code>fact_nav</code>	- Daily NAV snapshots (<code>fund_id</code> , <code>date_id</code> , <code>nav</code> , <code>daily_return</code>)
<code>fact_aum</code>	- Monthly AUM records (<code>fund_id</code> , <code>date_id</code> , <code>aum_cr</code>)
<code>fact_transactions</code>	- Investor transactions (<code>fund_id</code> , <code>date_id</code> , <code>amount</code> , <code>mode</code> , <code>age group</code> , <code>state</code>)
<code>fact_performance</code>	- Pre-computed performance metrics (CAGR, Sharpe, VaR, etc.)

Row Counts

fact_nav: ~73,000 NAV records across 40 schemes over 4 years

fact_transactions: ~33,000 transaction records with demographic metadata

fact_aum: ~90 AUM snapshots (monthly per fund house)

dim_fund: 40 scheme entries with category and risk classification

dim_date: ~1,460 date entries spanning 2022-2026

Data Quality Measures

- Forward-fill (ffill) for missing NAV values on weekends and public holidays.
- Unit suffix removal from columns (e.g., '6.05 L Cr' converted to 605000.0).
- Date range validation to ensure consistency across all fact tables.
- Foreign key integrity checks between dimension and fact tables.

3. Exploratory Data Analysis

The EDA notebook (notebooks/01_eda.ipynb) performs 10 analytical tasks covering NAV trends, AUM concentration, investor demographics, and transaction patterns. Key highlights are summarized below.

NAV Trends (2022-2026)

NAV trends across all 40 schemes show a consistent upward trajectory over the 4-year period. Large-cap and Flexi-cap funds display steady, low-volatility growth, while Small-cap and Mid-cap funds exhibit higher volatility with periodic drawdowns followed by sharp recoveries. The COVID recovery rally in early 2023 is visible across all categories.

AUM Concentration

The AUM distribution is heavily concentrated among the top fund houses. SBI Mutual Fund leads with approximately 6.05 lakh crore in AUM, followed by HDFC Mutual Fund and ICICI Prudential Mutual Fund. The top 3 fund houses account for over 55% of total industry AUM.

- SBI Mutual Fund: ~6.05 L Cr
- HDFC Mutual Fund: ~4.80 L Cr
- ICICI Prudential Mutual Fund: ~4.20 L Cr
- Nippon India Mutual Fund: ~3.10 L Cr
- Kotak Mahindra Mutual Fund: ~2.90 L Cr

Investor Demographics

Analysis of ~33,000 transaction records reveals the following demographic patterns:

- Age Group: The 36-45 age bracket is the most active, contributing ~32% of transaction volume.
- Payment Mode: UPI dominates with over 40% share, followed by Net Banking (28%) and NEFT (15%).
- Geographic Distribution: Maharashtra leads (18%), followed by Gujarat (14%) and Karnataka (12%).
- Top 3 states account for 44% of all transactions.

Folio Growth Trends

Folio (investor account) counts have grown steadily from ~38 lakh in 2022 to ~65 lakh in 2026, representing a CAGR of approximately 14%. This indicates strong retail investor participation in the mutual fund industry.

4. Performance Analytics

The Performance notebook (notebooks/02_performance_analytics.ipynb) computes a comprehensive set of risk-return metrics for all 40 schemes and compares them against the NIFTY 50 benchmark.

Metrics Computed

CAGR (1y / 3y / 5y)	- Annualized returns over multiple horizons
Sharpe Ratio	- Risk-adjusted return relative to risk-free rate ($R_f = 6.5\%$)
Sortino Ratio	- Downside-risk-adjusted return
Alpha	- Excess return over CAPM-expected return
Beta	- Systematic risk relative to NIFTY 50
Max Drawdown	- Peak-to-trough decline over the period
Tracking Error	- Standard deviation of active returns

Value-at-Risk (VaR) Analysis

Three VaR methodologies were implemented and compared:

- Parametric VaR (95% and 99% confidence) - assumes normal distribution of returns.
- Historical VaR - based on empirical return distribution percentiles.
- Conditional VaR (CVaR) - expected shortfall beyond the VaR threshold.

Small Cap funds consistently show the highest VaR values across all methods, with parametric VaR (95%) ranging from -3.5% to -5.2% daily. Large Cap funds have the lowest VaR, typically -1.8% to -2.5%.

Fund Scorecard (0-100)

A composite score (0-100) was developed to rank funds across multiple dimensions: returns (30%), risk-adjusted performance (25%), downside protection (20%), consistency (15%), and AUM stability (10%). The top-ranked funds and their scores are:

- Quant Small Cap Fund - Direct Growth: Score 88.5
- SBI Bluechip Fund - Direct Growth: Score 86.2
- HDFC Flexi Cap Fund - Direct Growth: Score 84.7
- Nippon India Small Cap Fund - Direct Growth: Score 82.1
- ICICI Pru Value Discovery Fund - Direct Growth: Score 80.9

Benchmark Comparison vs. NIFTY 50

The top 5 funds by score were compared against the NIFTY 50 benchmark on cumulative returns. All 5 funds outperformed the benchmark over the 4-year period, with Quant Small Cap Fund delivering ~185% cumulative returns vs NIFTY 50's ~72%. However, the outperformance comes with significantly higher volatility and drawdown.

5. Advanced Analytics

VaR Deep Dive & Backtesting

VaR estimates were backtested using the Kupiec Proportion of Failures (POF) test. For each fund, the actual number of daily returns exceeding the 95% VaR threshold was compared against the expected 5% failure rate.

Top 5 funds by Parametric VaR (95%):

- Quant Small Cap Fund: VaR -5.2% (Backtest: 4.8% failures - within acceptable range)
- Nippon India Small Cap Fund: VaR -4.9% (Backtest: 5.3% failures)
- SBI Small Cap Fund: VaR -4.7% (Backtest: 4.6% failures)
- Kotak Small Cap Fund: VaR -4.5% (Backtest: 5.1% failures)
- HDFC Mid-Cap Opportunities Fund: VaR -3.8% (Backtest: 4.9% failures)

Cohort Analysis - Investor Retention

A quarterly cohort retention heatmap was constructed using transaction data. Key findings include:

- Q1 2024 cohort shows the highest 3-month retention rate at 78%.
- Retention rates decline over subsequent quarters (average 62% by Q4 2025).
- Seasonal spikes in Q1 (Jan-Mar) across all years suggest year-start investment patterns.
- SIP investors show 15-20% higher retention compared to lump-sum investors.

Fund Recommender Engine

A rule-based recommender engine was developed to suggest funds based on investor risk profiles:

- Conservative: SBI Bluechip Fund, HDFC Flexi Cap Fund, ICICI Pru Balanced Advantage Fund
- Moderate: Mirae Asset Large & Midcap Fund, Kotak Equity Opportunities Fund, Axis Growth Fund
- Aggressive: Quant Small Cap Fund, Nippon India Small Cap Fund, SBI Small Cap Fund, Motilal Oswal Midcap Fund

The engine uses a scoring matrix that weighs past returns, volatility, drawdown, and category rank to match funds to the user's risk tolerance and investment horizon.

6. Dashboard Walkthrough

The Power BI dashboard comprises 4 interconnected pages with drill-through navigation and cross-filtering. Each page includes at least 2 slicers for interactive analysis.

Page 1: Industry Overview

This page provides a high-level industry snapshot with the following visual elements:

- KPI Cards: Total AUM, Total Folios, Avg NAV Growth, Total Transactions.
- NAV Trend: Line chart showing average NAV across all schemes over time.
- AUM by Fund House: Horizontal bar chart highlighting top fund houses.
- Folio Growth: Area chart tracking folio count growth from 2022-2026.
- Slicers: Fund Category (Large Cap, Small Cap, etc.), Date Range (slider).

Page 2: Fund Performance

Detailed performance analytics for individual funds:

- Fund Scorecard: Ranked fund list with composite scores (0-100).
- VaR Comparison: Table comparing Parametric, Historical, and CVaR across funds.
- Benchmark Comparison: Line chart of top 5 funds vs NIFTY 50 cumulative returns.
- Risk-Return Scatter: Scatter plot of CAGR vs Std Dev with fund labels.
- Slicers: Fund House, Risk Level, Benchmark toggle.

Page 3: Investor Analytics

Demographic and behavioral analysis of investors:

- Age Group Distribution: Donut chart of transaction volume by age bracket.
- Geographic Map: India map (choropleth) showing state-wise transaction volume.
- Transaction Trends: Stacked bar chart of monthly transactions by payment mode.
- Gender Analysis: Side-by-side comparison of investment amounts by gender.
- Slicers: State, Age Group, Payment Mode.

Page 4: SIP Trends

Systematic Investment Plan (SIP) inflow analysis:

- SIP Inflow Growth: Line chart tracking monthly SIP inflows over time.
- Category-wise SIP Inflows: Stacked area chart by fund category.
- Payment Mode Breakdown: Waterfall chart of SIP transactions by mode.
- Average SIP Ticket Size: KPI showing trend of average SIP amount.
- Slicers: Fund Category, SIP Tenure, Date Range.

7. Bonuses Implemented

B1: Cron-Ready ETL Scheduling

The ETL pipeline is designed for production scheduling. A cron expression (0 20 * * 1-5) schedules the pipeline to run at 8 PM on weekdays. The script checks for new data files, processes incrementally, and logs results to etl_pipeline.log. A batch/powershell wrapper (run_etl.bat) is provided for Windows Task Scheduler integration.

B2: Streamlit Web Application

A multi-page Streamlit application (streamlit_app/) provides interactive access to analytics:

- Overview Page: KPI cards, NAV trend chart, AUM bar chart.
- Screener Page: Filterable fund table with scorecard rankings.
- Performance Page: Individual fund performance metrics and charts.
- Optimizer Page: Markowitz Efficient Frontier with optimal portfolio suggestions.
- Monte Carlo Page: 5-year NAV simulation with confidence bands for selected funds.

B3: Monte Carlo Simulation

A Monte Carlo simulator projects NAV for 5 years (1,260 trading days) using Geometric Brownian Motion calibrated on historical log returns. Running 10,000 simulations, the model generates:

- Expected NAV path with 50th, 75th, 90th, and 95th percentile confidence bands.
- Probability distribution of terminal NAV after 5 years.
- Downside risk metrics: probability of loss, expected shortfall at 95% confidence.
- Interactive parameter controls (volatility, drift, time horizon) in the Streamlit app.

B4: Markowitz Efficient Frontier

Using `scipy.optimize.minimize` (SLSQP method), the optimizer computes the Efficient Frontier for any subset of funds. Key features:

- Max Sharpe portfolio: highest risk-adjusted return allocation among 5 selected funds.
- Min Volatility portfolio: lowest-risk allocation.
- Efficient Frontier curve plotted with 100 simulated portfolios.
- Constraints: full investment (sum weights = 1), no short selling (weights ≥ 0).

B5: HTML Email Report Generator

A weekly performance summary generator (scripts/email_report.py) produces styled HTML reports:

- Top/Bottom 5 performers table with weekly returns.
- Market overview section with benchmark comparison.
- Interactive charts rendered as embedded base64 PNG images.
- Responsive design compatible with major email clients (Gmail, Outlook).
- SMTP integration for automated delivery to subscriber list.

8. Conclusion

Key Deliverables Completed

- End-to-end ETL pipeline with automated scheduling.
- SQLite star-schema database with 6 tables and ~106K records.
- Comprehensive EDA covering NAV trends, AUM, demographics, and transactions.
- Risk-return analytics with CAGR, Sharpe, Sortino, Alpha, Beta, VaR, and CVaR.
- Fund scorecard ranking system (0-100) for 40 mutual fund schemes.
- 4-page interactive Power BI dashboard with cross-filtering and drill-through.
- 5 bonus features: cron ETL, Streamlit app, MC simulation, Markowitz optimizer, email reports.

Technologies Used

Python 3.11+	- pandas, numpy, scipy, yfinance, fpdf2, matplotlib, seaborn
SQLite 3	- star-schema database with indexed fact tables
Power BI Desktop	- 4-page dashboard with DAX measures and slicers
Streamlit	- Multi-page web app with interactive controls
Jupyter Lab	- 3 notebooks for exploratory and advanced analysis

Future Scope

- Real-time API integration with live NAV feeds from AMFI/mfapi.in.
- ML-based NAV prediction using LSTM or Prophet models.
- Natural Language Query (NLQ) interface for the dashboard.
- Portfolio rebalancing alerts based on drift from target allocation.
- Mobile app integration via Power BI embedded analytics.

This report is generated automatically as part of the Bluestock Mutual Fund Analytics Platform capstone project. All data sourced from AMFI India and mfapi.in public APIs.